FUNCTORIAL PROPERTIES OF NONSINGULAR PROJECTIVE MODELS UNDER FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the birational reduction with the author's marked detail, the normalization-properness argument for (1), the purely inseparable normal-regular-smooth chain for (2), and the global-section and base-change arguments for (3) and (4) are all retained verbatim.

Editorial note. The label lemma-smooth-models names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Normalizations of proper schemes. Varieties, Lemma varieties-lemma-normalization-locally-algebraic; Morphisms, Lemmas morphisms-lemma-finite-proper and morphisms-lemma-composition-proper; Varieties, Lemma varieties-lemma-dim-1-proper-projective. Makes the normalization of a proper curve finite hence proper, normal, irreducible and birational to the reduced fibre, identifying the smooth model.
- Improving models by purely inseparable change of fields. Varieties, Lemma varieties-lemma-finite-extension-geometrically-normal; Properties, Lemma properties-lemma-normal-dimension-1-regular; Varieties, Lemma varieties-lemma-geometrically-regular-smooth. Finds a finite purely inseparable K/k making the model geometrically normal, hence geometrically regular for curves, hence smooth.
- Global functions and base change. Varieties, Lemmas varieties-lemma-geometrically-integral and varieties-lemma-regular-functions-proper-variety; Morphisms, Lemma morphisms-lemma-base-change-smooth. Gives H^0 = K for geometrically reduced proper varieties and smoothness of base changes of smooth morphisms.

Source reference. Original lemma 01VB: sources/stacks-chow/tags.txt, source lines 2440–2440. Prerequisite pointer invoked at proof line(s) 538; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01W3: sources/stacks-chow/tags.txt, source lines 2468–2468. Prerequisite pointer invoked at proof line(s) 510; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 01WN: sources/stacks-chow/tags.txt, source lines 2488–2488. Prerequisite pointer invoked at proof line(s) 509; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BX2: sources/stacks-chow/tags.txt, source lines 14965–14965. Prerequisite pointer invoked at proof line(s) 522; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A26: sources/stacks-chow/tags.txt, source lines 12694–12694. Prerequisite pointer invoked at proof line(s) 514; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BXT: sources/stacks-chow/tags.txt, source lines 14991–14991. Prerequisite pointer invoked at proof line(s) 518; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 038K: sources/stacks-chow/tags.txt, source lines 4239–4239. Prerequisite pointer invoked at proof line(s) 528; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 038X: sources/stacks-chow/tags.txt, source lines 4252–4252. Prerequisite pointer invoked at proof line(s) 524; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BXR: sources/stacks-chow/tags.txt, source lines 14989–14989. Prerequisite pointer invoked at proof line(s) 508; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04L2: sources/stacks-chow/tags.txt, source lines 5993–5993. Prerequisite pointer invoked at proof line(s) 530; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a geometrically irreducible curve over k. For a field extension K/k denote Y_K a nonsingular projective model of $(X_K)_{red}$.*

(1) *If X is proper, then Y_K is the normalization of X_K.*
(2) *There exists K/k finite purely inseparable such that Y_K is smooth.*
(3) *Whenever Y_K is smooth*[1] *we have $H^0(Y_K, \mathcal{O}_{Y_K}) = K$.*
(4) *Given a commutative diagram*

$$\begin{array}{ccc} \Omega & \longleftarrow & K' \\ \uparrow & & \uparrow \\ K & \longleftarrow & k \end{array}$$

of fields such that Y_K and $Y_{K'}$ are smooth, then $Y_\Omega = (Y_K)_\Omega = (Y_{K'})_\Omega$.

Proof. Let X' be a nonsingular projective model of X. Then X' and X have isomorphic nonempty open subschemes. In particular X' is geometrically irreducible as X is (some details omitted). Thus we may assume that X is projective.

[1] Or even geometrically reduced.

Assume X is proper. Then X_K is proper and hence the normalization $(X_K)^\nu$ is proper as a scheme finite over a proper scheme (Varieties, Lemma 0BXR and Morphisms, Lemmas 01WN and 01W3). On the other hand, X_K is irreducible as X is geometrically irreducible. Hence X_K^ν is proper, normal, irreducible, and birational to $(X_K)_{red}$. This proves (1) because a proper curve is projective (Varieties, Lemma 0A26).

Proof of (2). As X is proper and we have (1), we can apply Varieties, Lemma 0BXT to find K/k finite purely inseparable such that Y_K is geometrically normal. Then Y_K is geometrically regular as normal and regular are the same for curves (Properties, Lemma 0BX2). Then Y is a smooth variety by Varieties, Lemma 038X.

If Y_K is geometrically reduced, then Y_K is geometrically integral (Varieties, Lemma 038K) and we see that $H^0(Y_K, \mathcal{O}_{Y_K}) = K$ by Varieties, Lemma 04L2. This proves (3) because a smooth variety is geometrically reduced (even geometrically regular, see Varieties, Lemma 038X).

If Y_K is smooth, then for every extension Ω/K the base change $(Y_K)_\Omega$ is smooth over Ω (Morphisms, Lemma 01VB). Hence it is clear that $Y_\Omega = (Y_K)_\Omega$. This proves (4). □